Why callbacks

Nonlinear devices expose i(v) and g=di/dv to the MNA Newton loop

The idea

- GOLDEN: diode_loop_newton converges; callback returns i and g at vd

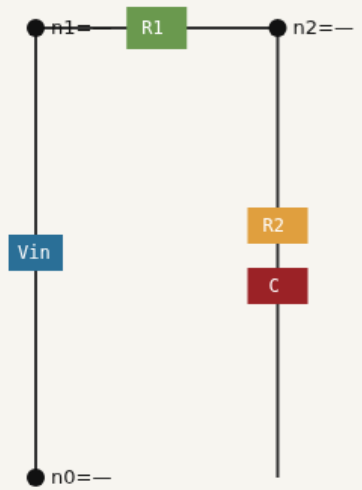
i(v)

DEVICE CALLBACKS

i(v)

Step 1 / 5 · i · module03-03-device-callbacks

elements=5



Explanation

Diode current Is(exp(v/Vt)-1).

- i

```
exp
```

i(v)

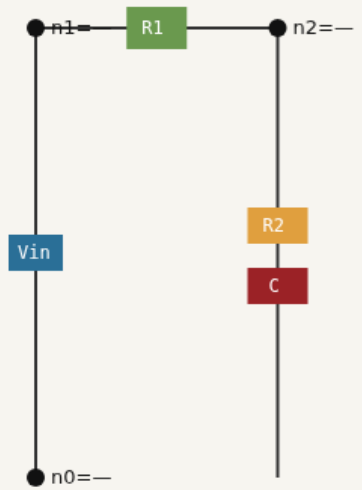
g=di/dv

DEVICE CALLBACKS

g=di/dv

Step 2 / 5 · g · module03-03-device-callbacks

elements=5



Explanation

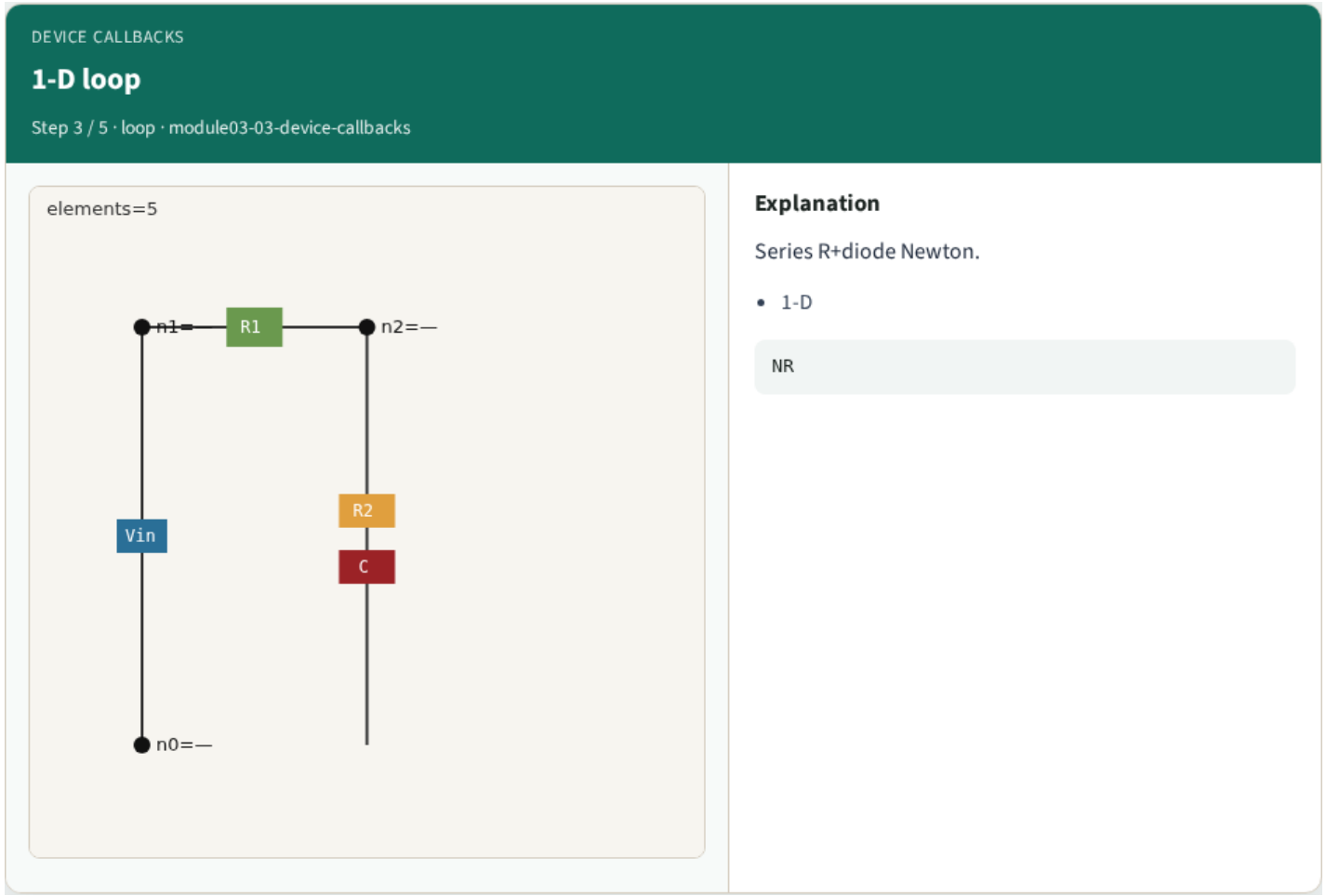
Conductance for Jacobian.

- g

```
Is/Vt·exp
```

g=di/dv

1-D loop



1-D loop

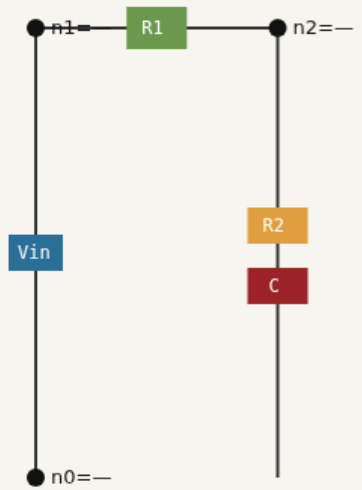
Converge

DEVICE CALLBACKS

Converge

Step 4 / 5 · cv · module03-03-device-callbacks

conv=true iters=25



Explanation

diode_loop_newton converges.

- ok

```
true
```

Converge

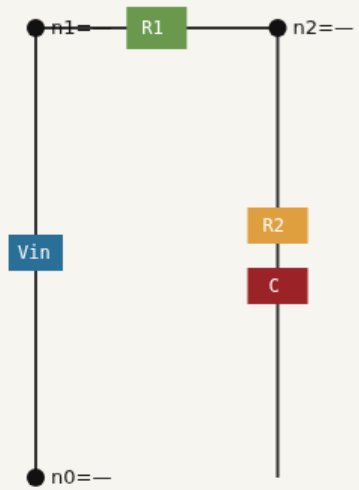
Hook

DEVICE CALLBACKS

Hook

Step 5 / 5 · hook · module03-03-device-callbacks

callbacks



Explanation

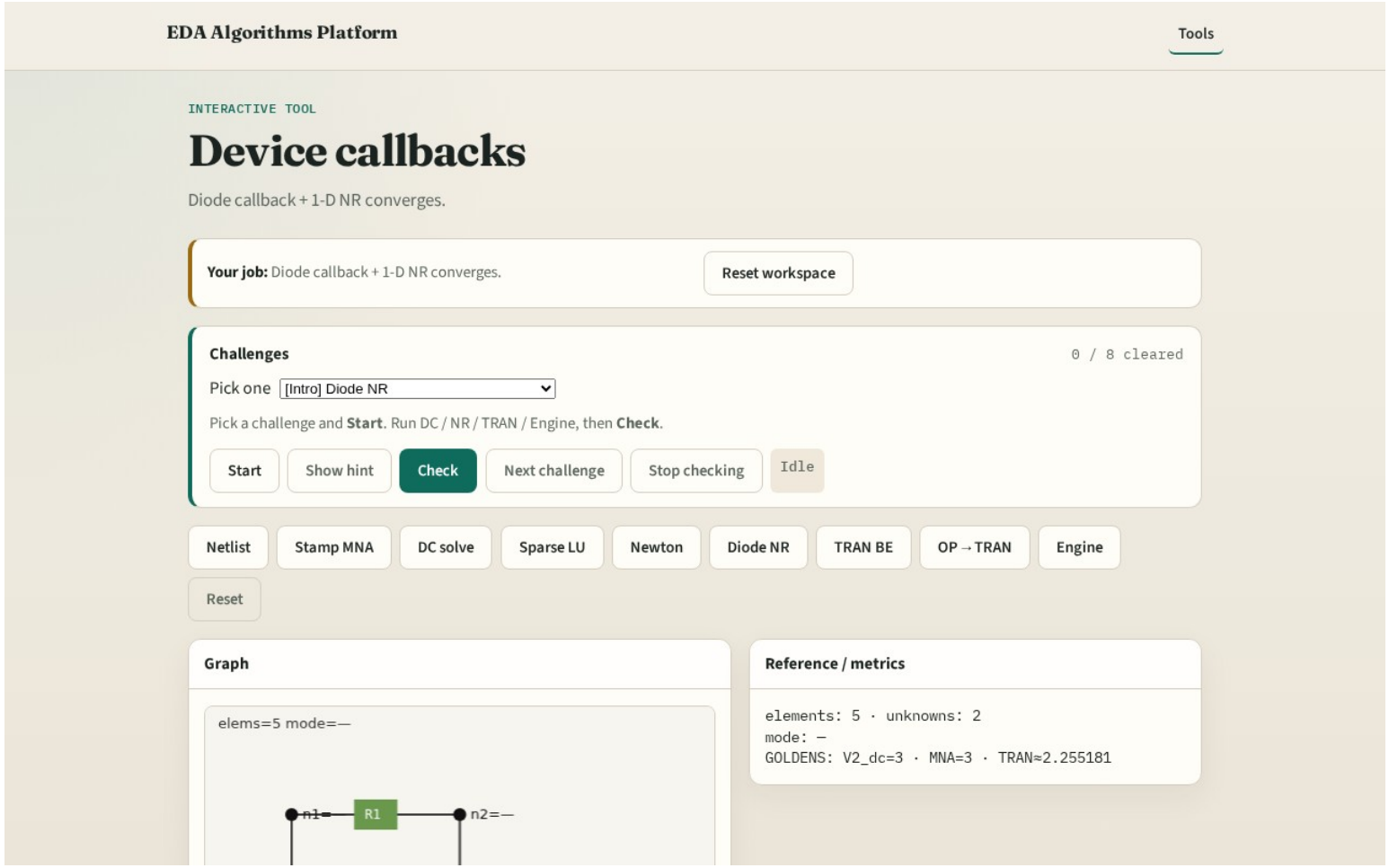
Callbacks plug into MNA stamps.

- API

```
i,g
```

Hook

Browser lab track



Browser lab starter

Implement track

- Call device_callback_diode(0.0) and diode_loop_newton(); assert converged

Pitfalls

- Using linear diode model
- Forgetting companion conductance in transient

Your turn

- Next: backward-Euler transient on RC